Unit 8 Progress Check: MCQ

Buffer solutions containing Na_2CO_3 and $NaHCO_3$ range in pH from 10.0 to 11.0. The chemical equation below represents the equilibrium between $CO_3{}^{2-}$ and H_2O, and the table lists the composition of four different buffer solutions at $25^\circ C$.

$$CO_3{}^{2-}(aq) + H_2O(l) \rightleftarrows HCO_3{}^{-}(aq) + OH^{-}(aq) \qquad K_b = 2.1 \times 10^{-4} \text{ at } 25^\circ C$$

Buffer	$[NaHCO_3]$	$[Na_2CO_3]$	pH
1	0.150	0.100	?
2	0.200	0.200	10.32
3	0.100	0.100	10.32
4	0.100	0.200	?

1. Which of the following chemical equilibrium equations best shows what happens in the buffer solutions to minimize the change in pH when a small amount of a strong base is added?

(A) $H_3O^{+}(aq) + OH^{-}(aq) \rightleftarrows 2\,H_2O(l)$

(B) $HCO_3{}^{-}(aq) + OH^{-}(aq) \rightleftarrows CO_3{}^{2-}(aq) + H_2O(l)$

(C) $CO_3{}^{2-}(aq) + H_3O^{+}(aq) \rightleftarrows HCO_3{}^{-}(aq) + H_2O(l)$

(D) $CO_3{}^{2-}(aq) + H_2O(l) \rightleftarrows HCO_3{}^{-}(aq) + OH^{-}(aq)$

2. Which of the following mathematical expressions can be used to determine the approximate pH of buffer 1 ?

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(A) $\text{pH} = -\log(2.1 \times 10^{-4}) + \log\left(\frac{0.100}{0.150}\right) = 3.50$

(B) $\text{pH} = -\log(2.1 \times 10^{-4}) + \log\left(\frac{0.150}{0.100}\right) = 3.85$

(C) $\text{pH} = [14.00 + \log(2.1 \times 10^{-4})] + \log\left(\frac{0.100}{0.150}\right) = 10.15$

(D) $\text{pH} = [14.00 + \log(2.1 \times 10^{-4})] + \log\left(\frac{0.150}{0.100}\right) = 10.50$

---

3. Which mathematical expression can be used to explain why buffer 2 and buffer 3 have the same pH?

(A) $\log\left(\frac{0.200}{0.200}\right) = \log\left(\frac{0.100}{0.100}\right) = \log(1)$

(B) $-\log(K_w) - [-\log(K_b)] = \text{p}K_a$

(C) $0.200\ M = 2 \times (0.100\ M)$

(D) $0.200\ M - 0.200\ M = 0.100\ M - 0.100\ M$

4. $H_2O(l) + H_2O(l) \rightleftarrows H_3O^+(aq) + OH^-(aq) \qquad K_w = 7.0 \times 10^{-14}$ at $55^\circ C$

Pure water autoionizes as shown in the equation above. Based on this information, which of the following is correct?

(A) The autoionization equilibrium for pure water favors the formation of reactants at $55^\circ C$ compared to $25^\circ C$.

(B) The autoionization equilibrium for pure water produces the same amount of OH^- ions at $55^\circ C$ and $25^\circ C$.

(C) At $55^\circ C$, $\text{pH} = -\log\left(\sqrt{K_w}\right)$ for pure water.

(D) At $55^\circ C$, $\text{pH} = -\log(K_w)$ for pure water.

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Unit 8 Progress Check: MCQ

5. $K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$ at $25^\circ C$

Based on the information above, which of the following is true for a sample of pure water at $25^\circ C$?

(A) $[H_3O^+] = 7.0\ M$

(B) $[OH^-] = 1.0 \times 10^{-14}\ M$

(C) $pH = 10^{-7}$

(D) $pOH = 7.00$

6. $2\ H_2O(l) \rightleftarrows H_3O^+(aq) + OH^-(aq)\ \Delta H^\circ = +56\ kJ/mol_{rxn}$

The endothermic autoionization of pure water is represented by the chemical equation shown above. The pH of pure water is measured to be 7.00 at $25.0^\circ C$ and 6.02 at $100.0^\circ C$. Which of the following statements best explains these observations?

(A) At the higher temperature water dissociates less, $[H_3O^+] < [OH^-]$, and the water becomes basic.

(B) At the higher temperature water dissociates less, $[H_3O^+] = [OH^-]$, and the water remains neutral.

(C) At the higher temperature water dissociates more, $[H_3O^+] > [OH^-]$, and the water becomes acidic.

(D) At the higher temperature water dissociates more, $[H_3O^+] = [OH^-]$, and the water remains neutral.

7. Which of the following gives the best estimate for the pH of a $5 \times 10^{-4}\ M\ Sr(OH)_2(aq)$ solution at $25^\circ C$?

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(A) pH $\approx$ 3.0 because $Sr(OH)_2$ is a strong acid.

(B) pH $\approx$ 5.0 because $Sr(OH)_2$ is a weak acid.

(C) pH $\approx$ 9.0 because $Sr(OH)_2$ is a weak base.

(D) pH $\approx$ 11.0 because $Sr(OH)_2$ is a strong base.

8. Which of the following gives the best estimate for the pH of a 1×10^{-5} M $HClO_4(aq)$ solution at 25°C?

(A) pH $\approx$ 1.0 because $HClO_4$ is a strong acid.

(B) pH $\approx$ 5.0 because $HClO_4$ is a strong acid.

(C) pH $\approx$ 7.0 because ClO_4^- is a strong base.

(D) pH $\approx$ 9.0 because ClO_4^- is a strong base.

9. Which of the following is the correct mathematical relationship to use to calculate the pH of a 0.10 M aqueous HBr solution?

(A) $\text{pH} = [H_3O^+] = 0.10$

(B) $\text{pH} = -\log(1.0 \times 10^{-1})$

(C) $\text{pH} = 7.00 - (0.10)$

(D) $\text{pH} = \log(1.0 \times 10^{-1})$

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Unit 8 Progress Check: MCQ

10. $HCOOH(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + HCOO^-(aq)$ $K_a = 1.8 \times 10^{-4}$ at $25^\circ C$

Initial [HCOOH]	Percent Ionization	Equilibrium $[H_3O^+]$
0.150	3.4%	5.1×10^{-3}
0.100	4.2%	4.2×10^{-3}
0.0500	5.8%	2.9×10^{-3}

The equilibrium for the acid ionization of HCOOH is represented by the equation above and the table gives the percent ionization for HCOOH at different initial concentrations of the weak acid at $25^\circ C$. Based on the information, which of the following is true for a $0.125\ M$ aqueous solution of HCOOH?

(A) It has a larger percent ionization, a lower $[H_3O^+]_{eq}$, and a lower pH than a $0.150\ M$ HCOOH solution does.

(B) It has a larger percent ionization, a lower $[H_3O^+]_{eq}$, and a higher pOH than a $0.150\ M$ HCOOH solution does.

(C) It has a lower percent ionization, a larger $[H_3O^+]_{eq}$, and a higher pOH than a $0.100\ M$ HCOOH solution does.

(D) It has a lower percent ionization, a larger $[H_3O^+]_{eq}$, and a higher pH than a $0.100\ M$ HCOOH solution.

11. $NH_3(aq) + H_2O(l) \rightleftarrows NH_4^+(aq) + OH^-(aq)$ $K_b = 1.8 \times 10^{-5}$ at $25^\circ C$

The reaction between NH_3 and water is represented above. A solution that is initially $0.200\ M$ in NH_3 has a pH of 11.28. Which of the following correctly predicts the pH of a solution for which $[NH_3]_{initial} = 0.100\ M$, and why?

(A) The pH will be lower than 11.28 because decreasing the initial concentration of NH_3 increases the equilibrium concentration of the conjugate acid NH_4^+.

(B) The pH will be lower than 11.28 because the equilibrium concentration of OH^- ions decreases when the initial concentration of the base decreases.

(C) The pH will be higher than 11.28 because decreasing the initial concentration of NH_3 decreases the equilibrium concentration of the conjugate acid NH_4^+.

(D) The pH will be higher than 11.28 because the equilibrium concentration of OH^- ions increases when the initial concentration of the base decreases.

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Unit 8 Progress Check: MCQ

12. $HClO(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + ClO^-(aq)\ K_a = 3.98 \times 10^{-8}$

The acid ionization equilibrium for $HClO$ at $25^\circ C$ is shown above. A $0.100\ M$ aqueous solution of this acid has a pH of about 4.2. If a solution with an initial concentration of $HClO$ of $0.200\ M$ is allowed to reach equilibrium at the same temperature, which of the following correctly predicts its pH, and why?

(A) The pH will be higher than 4.2 because increasing the concentration of the weak acid decreases the percent ionization.

(B) The pH will be higher than 4.2 because, for weak acid solutions, the pH is directly proportional to the initial concentration of the weak acid.

(C) The pH will be lower than 4.2 because increasing the concentration of the weak acid produces more H_3O^+ to establish equilibrium.

(D) The pH will be lower than 4.2 because the K_a of the weak acid is inversely proportional to the initial concentration of the weak acid.

13. The weak acid CH_3COOH has a pK_a of 4.76. A solution is prepared by mixing $500.\ mL$ of $0.150\ M\ CH_3COOH(aq)$ and $0.0200\ mol$ of $NaOH(s)$. Which of the following can be used to calculate the pH of the solution?

(A) $pH = 4.76 + \log(\frac{0.0200}{0.150}) = 3.88$

(B) $pH = 4.76 + \log(\frac{0.0200}{0.0750}) = 4.19$

(C) $pH = 4.76 + \log(\frac{0.0200}{0.0550}) = 4.32$

(D) $pH = 4.76 + \log(\frac{0.0950}{0.0750}) = 4.86$

14. Which of the following provides the correct mathematical expression to calculate the pH of a solution made by mixing $10.0\ ml$ of $1.00\ M$ HCl and $11.0\ mL$ of $1.00\ M$ $NaOH$ at $25^\circ C$?

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Unit 8 Progress Check: MCQ

(A) $\mathrm{pH} = -\log(0.0010) = 3.00$

(B) $\mathrm{pH} = 14.00 + \log(0.001) = 11.00$

(C) $\mathrm{pH} = 14.00 + \log\left(\frac{0.0010}{0.0210}\right) = 12.68$

(D) $\mathrm{pH} = 14.00 + \log\left(\frac{0.0010}{0.0110}\right) = 12.96$

15.

Substance	$\mathrm{p}K_a$
$NH_4{}^+(aq)$	9.25
$HCl(aq)$	< 1

A buffer solution is formed by mixing equal volumes of $0.12\ M\ NH_3(aq)$ and $0.10\ M\ HCl(aq)$, which reduces the concentration of both solutions by one half. Based on the $\mathrm{p}K_a$ data given in the table, which of the following gives the pH of the buffer solution?

(A) $\mathrm{pH} = -\log(0.050) = 1.30$

(B) $\mathrm{pH} = 9.25 + \log\left(\frac{0.010}{0.050}\right) = 8.55$

(C) $\mathrm{pH} = 9.25 + \log\left(\frac{0.060}{0.050}\right) = 9.32$

(D) $\mathrm{pH} = 14.00 - (-\log(0.010)) = 12.00$

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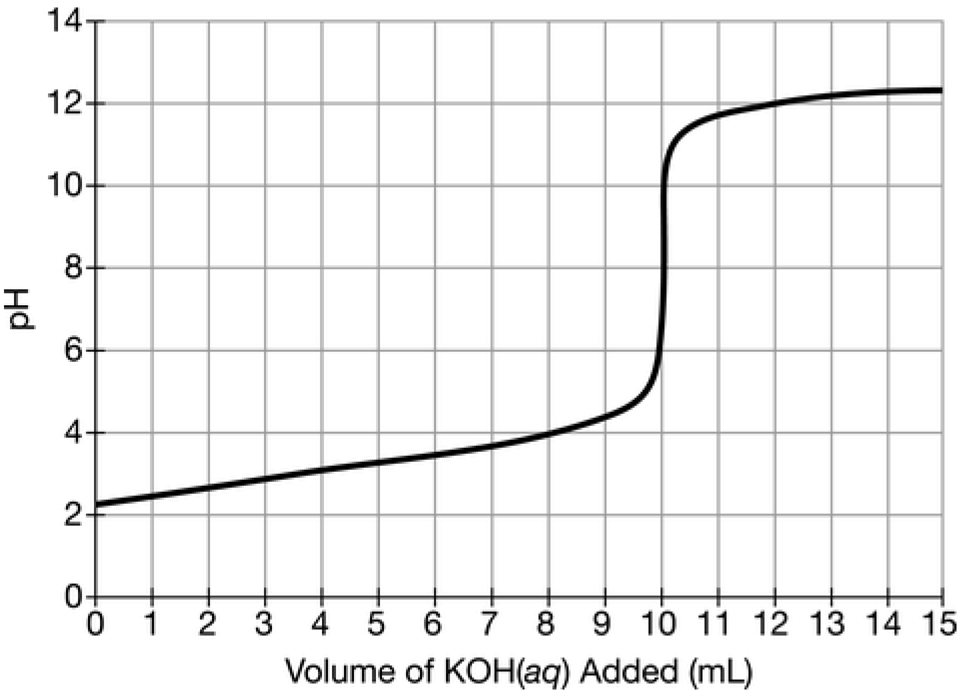
Unit 8 Progress Check: MCQ

16.



The pH versus volume data for the titration of $0.10\ M\ HNO_2(aq)$ with $0.10\ M\ KOH(aq)$ is plotted on the graph above. Based on the data, which of the following species is present in the greatest concentration after $6.0\ mL$ of $KOH(aq)$ has been added to the solution of $HNO_2(aq)$?

(A) $H^+(aq)$

(B) $HNO_2(aq)$

(C) $NO_2^-(aq)$

(D) $OH^-(aq)$

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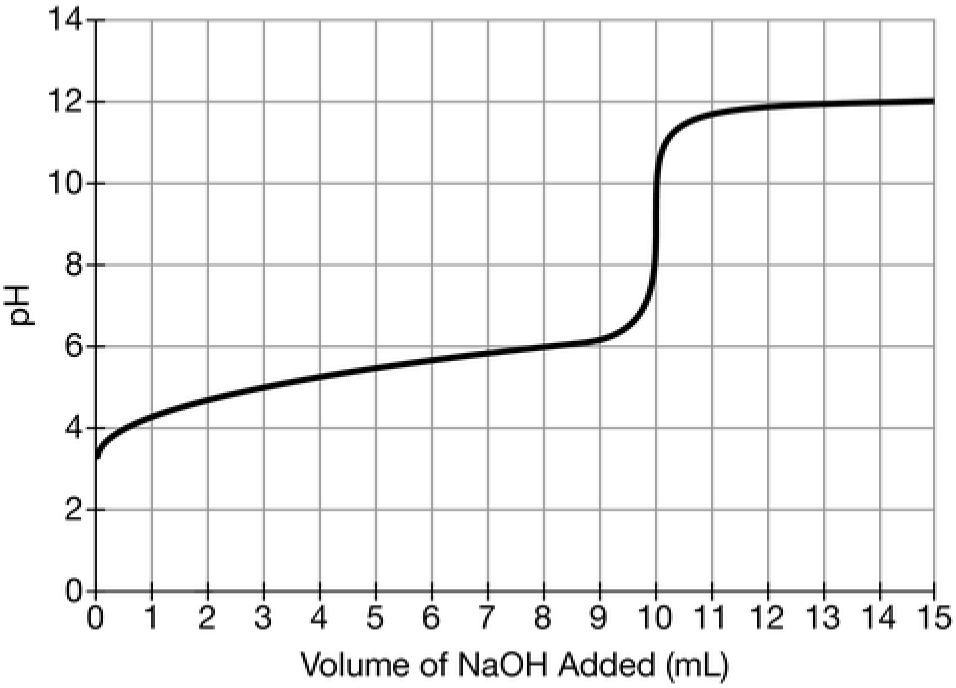
Unit 8 Progress Check: MCQ

17.



A $0.20\ M$ solution of the weak acid potassium hydrogen phthalate (KHP) is titrated with $0.10\ M$ $\text{NaOH}(aq)$. Based on the titration curve shown in the graph above, the $\text{p}K_a$ of KHP is closest to which of the following?

(A) 3.6

(B) 5.4

(C) 8.8

(D) 11.8

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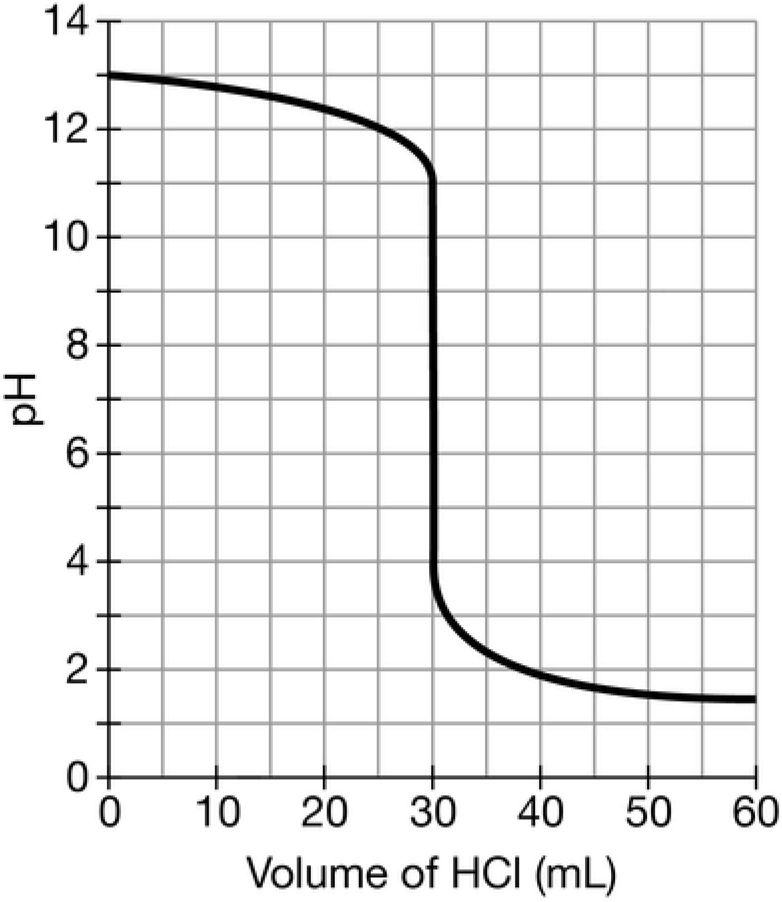
Unit 8 Progress Check: MCQ

18.



A $60.$ mL sample of $NaOH(aq)$ was titrated with 0.10 M $HCl(aq)$. Based on the resulting titration curve shown above, what was the approximate concentration of $NaOH$ in the sample?

(A) 0.033 M

(B) 0.050 M

(C) 0.10 M

(D) 0.20 M

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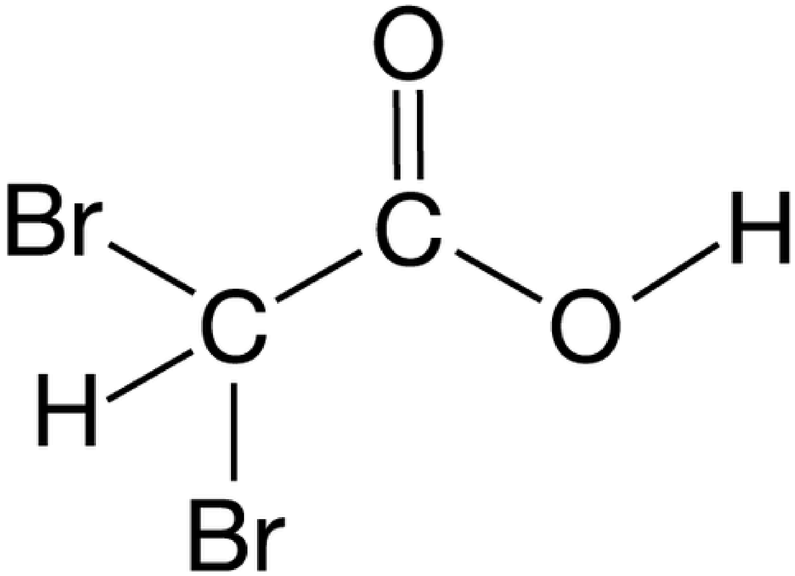
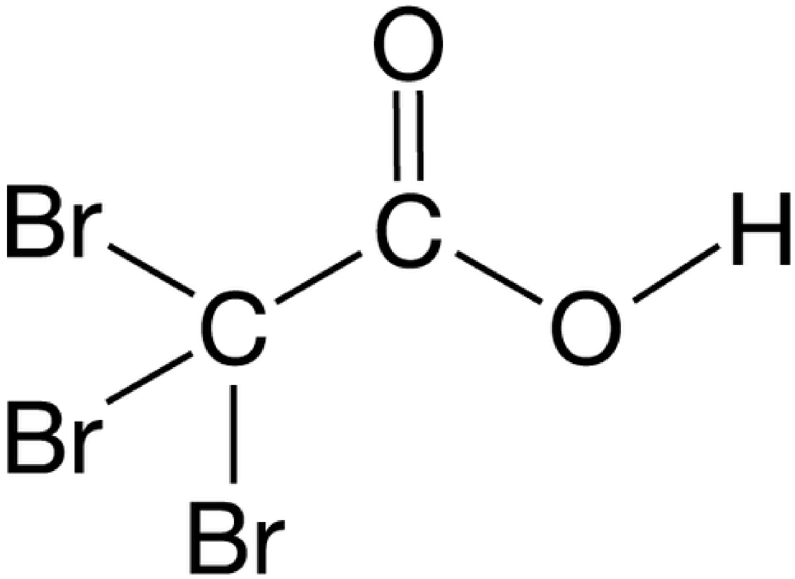
Unit 8 Progress Check: MCQ

19.

Name	Structure	K_a (at 25 °C)
Acid 1		3.3×10^{-2}
Acid 2		1.9×10^{-1}

The table above provides information on two weak acids. Which of the following explains the difference in their acid strength?

(A) Acid 1 is a stronger acid because it has more acidic hydrogen atoms than acid 2.

(B) Acid 1 is a stronger acid because it can make more hydrogen bonds than acid 2.

(C) Acid 2 is a stronger acid because it is a larger, more polarizable molecule with stronger intermolecular forces than acid 1.

(D) Acid 2 is a stronger acid because it has a more stable conjugate base than acid 1 due to the greater number of electronegative Br atoms.

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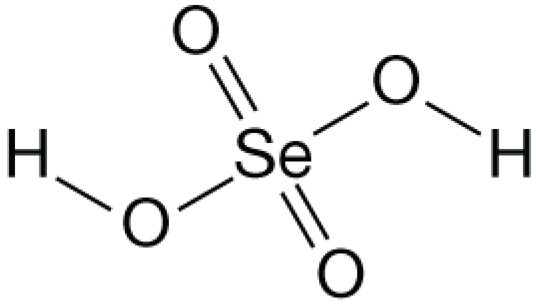
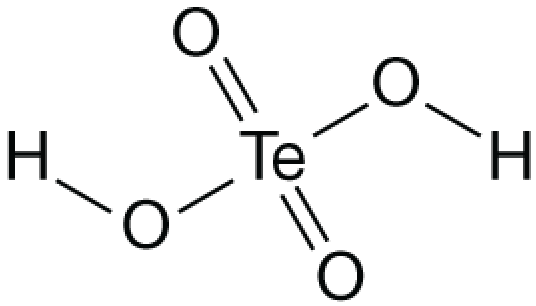
Unit 8 Progress Check: MCQ

20. $H_2XO_4(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + HXO_4^-(aq)\ K_{a1}$

$HXO_4^-(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + XO_4^{2-}(aq)\ K_{a2}$

Acid	K_{a1}	K_{a2}
	$>> 1$	2×10^{-2}
	2×10^{-8}	1×10^{-11}

The equilibrium reactions for diprotic oxoacids with a general formula H_2XO_4 are represented by the equations above. The acid ionization constants for H_2SeO_4 and H_2TeO_4 are provided in the table. Which of the following best explains the difference in strength for these two acids?

(A) H_2SeO_4 is weaker because Se has a smaller positive formal charge than Te, resulting in a decrease in its ability to transfer an H^+ to H_2O.

(B) H_2TeO_4 is weaker because Te has a smaller positive formal charge than Te, resulting in a decrease in its ability to transfer an H^+ to H_2O.

(C) H_2SeO_4 is weaker because Se is more electronegative than Te, resulting in more stable conjugate bases $HSeO_4^-$ and SeO_4^{2-} than those for H_2TeO_4 .

(D) H_2TeO_4 is weaker because Te is less electronegative than Se, resulting in less stable conjugate bases $HTeO_4^-$ and TeO_4^{2-} than those for H_2SeO_4.

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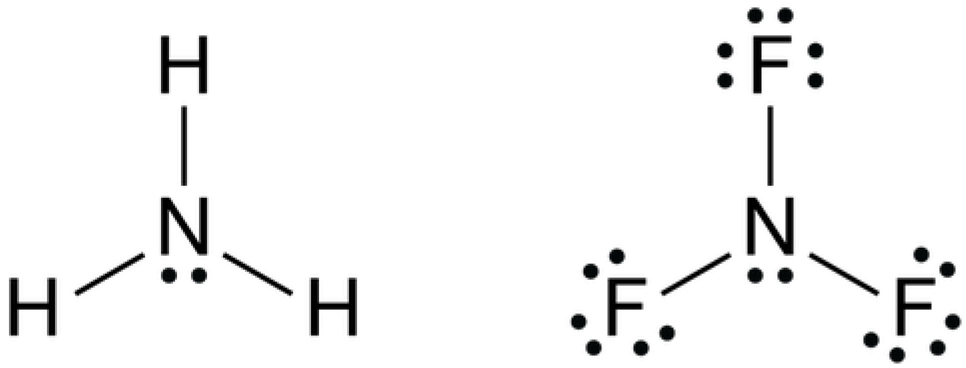
Unit 8 Progress Check: MCQ

21.



Lewis diagrams of the weak bases NH_3 and NF_3 are shown above. Based on these diagrams, which of the following predictions of their relative base strength is correct, and why?

(A) NF_3 is a stronger base than NH_3 because more than one resonance structure exists for its conjugate acid NF_3H^+, making it more stable than $NH_4{}^+$.

(B) NF_3 is a stronger base than NH_3 because of the greater electronegativity of F compared with H.

(C) NF_3 is a weaker base than NH_3 because of the greater electronegativity of F compared with H.

(D) NF_3 is a weaker base than NH_3 because more than one resonance structure exists for its conjugate acid NF_3H^+, making it more stable than $NH_4{}^+$.

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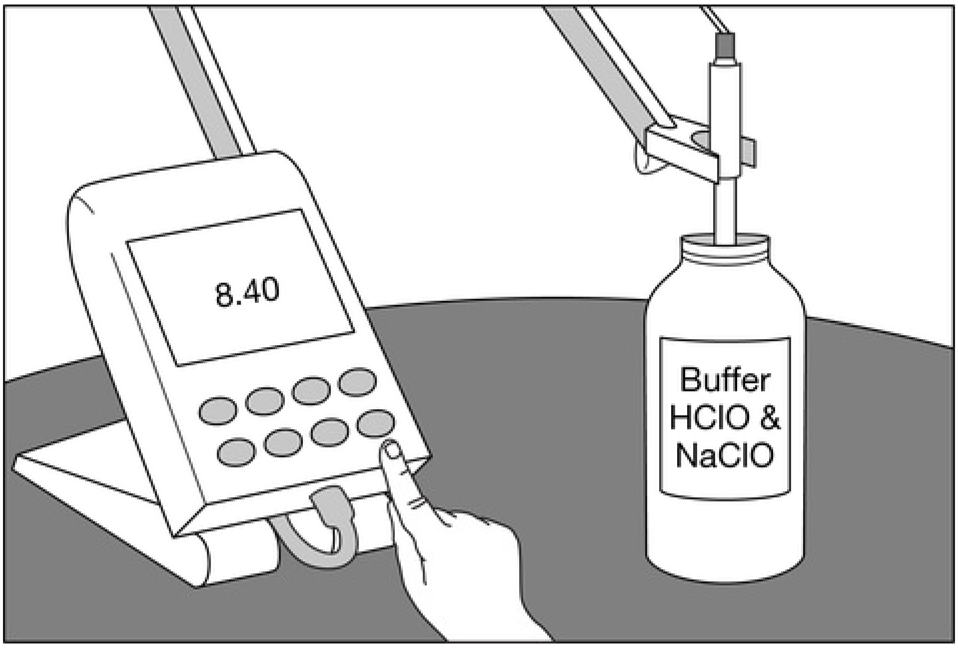
Unit 8 Progress Check: MCQ

22.



A student measures the pH of a $0.0100\ M$ buffer solution made with $HClO$ and $NaClO$, as shown above. The pK_a of $HClO$ is 7.40 at $25^\circ C$. Based on this information, which of the following best compares the relative concentrations of ClO^- and $HClO$ in the buffer solution?

(A) $[ClO^-] = [HClO]$

(B) $[ClO^-] > [HClO]$

(C) $[ClO^-] < [HClO]$

(D) It is not possible to compare the concentrations without knowing the pK_b of ClO^-.

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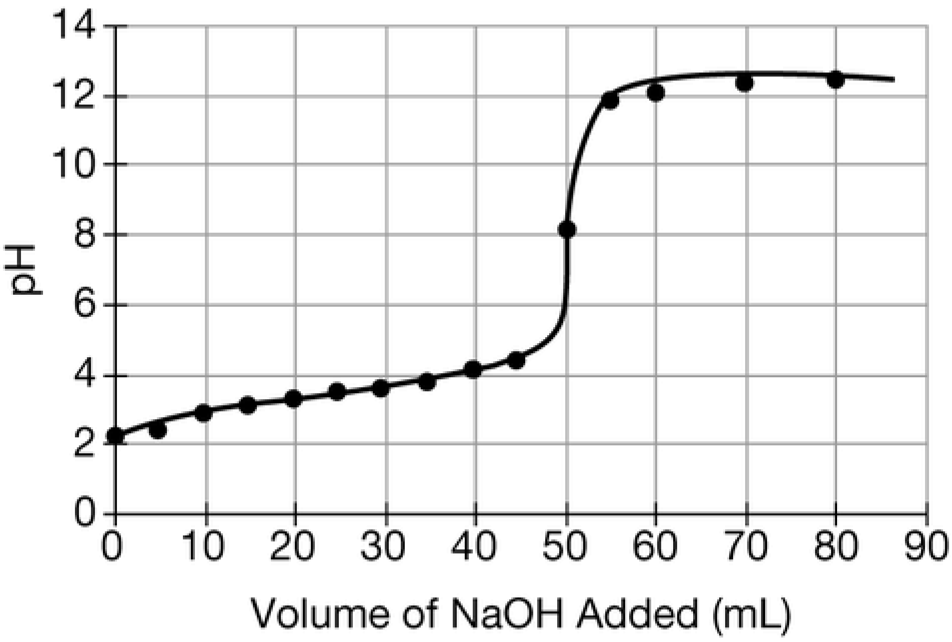
Unit 8 Progress Check: MCQ

23.



HCNO is a weak acid with a pK_a value of 3.5. The graph above shows the results of a titration of an aqueous solution of HCNO with 0.100 M NaOH. Based on the results, the concentration of CNO^- is greater than the concentration of HCNO at which of the following pH values?

(A) 2.0

(B) 3.0

(C) 3.5

(D) 4.0

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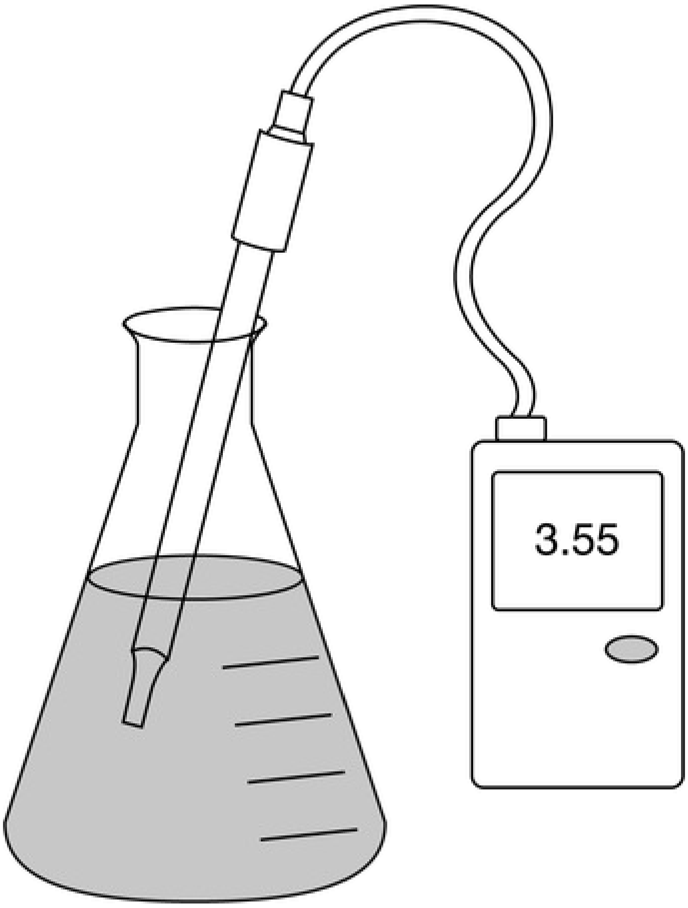
Unit 8 Progress Check: MCQ

24.



$C_2H_3COOH(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + C_2H_3COO^-(aq)$ $pK_a = 4.25$

The weak acid ionization equilibrium for C_2H_3COOH is represented by the equation above. A student measures the pH of $C_2H_3COOH(aq)$ using a probe and a pH meter in the experimental setup shown. Based on the information given, which of the following is true?

(A) $[C_2H_3COOH] > [C_2H_3COO^-]$ since the pK_a of the weak acid is less than pK_w.

(B) $[C_2H_3COOH] > [C_2H_3COO^-]$ since the pK_a of the weak acid is greater than the pH of the solution.

(C) $[C_2H_3COOH] < [C_2H_3COO^-]$ since the pK_b for $C_2H_3COO^-$ is less than pK_w.

(D) $[C_2H_3COOH] < [C_2H_3COO^-]$ since the K_b for $C_2H_3COO^-$ is less than the K_a for C_2H_3COOH.

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Unit 8 Progress Check: MCQ

25. $CH_3CH_2COOH(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + CH_3CH_2COO^-(aq)$ $K_a = 1.4 \times 10^{-5}$ at $25^\circ C$
$CH_3CH_2COO^-(aq) + H_2O(l) \rightleftarrows CH_3CH_2COOH(aq) + OH^-(aq)$ $K_b = 7.4 \times 10^{-10}$ at $25^\circ C$

The acid equilibrium for CH_3CH_2COOH and the base equilibrium for $CH_3CH_2COO^-$ are represented above. One liter of a buffer solution with $pH = 4.85$ is made by mixing $0.100\ M$ CH_3CH_2COOH and $0.100\ M$ $NaCH_3CH_2COO$. If 10.0 mL of $0.500\ M$ NaOH is added to the buffer, which of the following is most likely the resulting pH, and why?

(A) The pH will be much less than 4.85, because the buffer will respond to the addition of NaOH by producing more CH_3CH_2COOH.

(B) The pH will be much greater than 4.85, because there is a large increase in the concentration of the weak base, $CH_3CH_2COO^-$.

(C) The pH will be slightly less than 4.85, because the addition of NaOH reduces the autoionization of H_2O.

(D) The pH will be slightly greater than 4.85, because some CH_3CH_2COOH will react with the added bases, resulting in a slight decrease in $[H_3O^+]$.

26. A buffer solution that is $0.100\ M$ in both HCOOH and HCOOK has a $pH = 3.75$. A student says that if a very small amount of $0.100\ M$ HCl is added to the buffer, the pH will decrease by a very small amount. Which of the following best supports the student's claim?

(A) $HCOO^-$ will accept a proton from HCl to produce more HCOOH and H_2O.

(B) HCOOH will accept a proton from HCl to produce more $HCOO^-$ and H_2O.

(C) $HCOO^-$ will donate a proton to HCl to produce more HCOOH and H_2O.

(D) HCOOH will donate a proton to HCl to produce more $HCOO^-$ and H_2O.

27. $CH_3COOH(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + CH_3COO^-(aq)$ $pK_a = 4.76$ at $25^\circ C$

The equilibrium representing the acid dissociation of CH_3COOH is shown above. A buffer solution is prepared by adding 0.10 mol of $NaOH(s)$ to 1.00 L of $0.30\ M$ CH_3COOH. Assuming the change in volume is negligible, which of the following expressions will give the pH of the resulting buffer at $25^\circ C$?

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(A) $\text{pH} = 4.76 + \log\frac{0.10}{0.30}$

(B) $\text{pH} = 4.76 + \log\frac{0.10}{0.20}$

(C) $\text{pH} = 4.76 + \log\frac{0.20}{0.10}$

(D) $\text{pH} = 4.76 + \log\frac{0.30}{0.10}$

28. To prepare a buffer solution for an experiment, a student measured out 53.49 g of $NH_4Cl(s)$ (molar mass 53.49 g/mol) and added it to 1.0 L of 1.0 M $NH_3(aq)$. However, in the process of adding the $NH_4Cl(s)$ to the $NH_3(aq)$, the student spilled some of the $NH_4Cl(s)$ onto the bench top. As a result, only about $50.$ g of $NH_4Cl(s)$ was actually added to the 1.0 M $NH_3(aq)$. Which of the following best describes how the buffer capacity of the solution is affected as a result of the spill?

(A) The solution has a greater buffer capacity for the addition of base than for acid, because $[NH_3] < [NH_4{}^+]$.

(B) The solution has a greater buffer capacity for the addition of base than for acid, because $[NH_3] > [NH_4{}^+]$.

(C) The solution has a greater buffer capacity for the addition of acid than for base, because $[NH_3] < [NH_4{}^+]$.

(D) The solution has a greater buffer capacity for the addition of acid than for base, because $[NH_3] > [NH_4{}^+]$.

29. $HC_3H_5O_2(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + C_3H_5O_2{}^-(aq)$ $\text{p}K_a = 4.87$

The acid ionization equilibrium for $HC_3H_5O_2$ is represented by the equation above. A mixture of 1.00 L of 0.100 M $HC_3H_5O_2$ and 0.500 L of 0.100 M NaOH will produce a buffer solution with a $\text{pH} = 4.87$. If the NaOH solution was mislabeled and was 1.00 M instead of 0.100 M, which of the following would be true?

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Unit 8 Progress Check: MCQ

(A) The pH of the resulting solution would still be 4.87 because buffer solutions regulate changes in pH regardless of how much NaOH is added.

(B) The pH of the resulting solution would be somewhat higher than 4.87 because adding NaOH that is 10 times more concentrated makes the concentration of the conjugate base 10 times larger.

(C) The pH of the resulting solution would be much higher than 4.87 because the weak acid would be completely neutralized by the larger amount of NaOH added.

(D) The pH of the resulting solution would still be 4.87 because the solution contains a large volume of the weak acid to react with the NaOH and maintain the same pH.

30. $HNO_2(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + NO_2{}^-(aq) \qquad pK_a = 3.25$

Sample	Solution Composition	pH
1	25.0 mL buffer solution	3.26
2	25.0 mL buffer solution	3.25
3	25.0 mL buffer solution and one drop of HCl	3.03
4	25.0 mL buffer solution	3.21
5	25.0 mL buffer solution	3.25

The acid ionization equilibrium for HNO_2 is represented by the equation above. A 250.0 mL buffer solution is prepared by mixing 125.0 mL of 0.20 M HNO_2 and 125.0 mL of 0.1 M NaOH. To test the buffer capacity, the pH is measured and recorded in the table for four samples of the buffer and one sample of a mixture of the buffer and HCl. Which of the following best helps explain why the pH of sample 4 is lower than the pH of the other samples containing only buffer solution?

(A) Prior to measuring the pH of sample 4, some water evaporated, resulting in an increase in the concentration of HNO_2 in the sample.

(B) After measuring the pH of the more acidic sample 3, the pH probe was not rinsed and wiped, resulting in the neutralization of a very small amount of the conjugate base in sample 4.

(C) Prior to measuring the pH of sample 4, the room temperature dropped suddenly, resulting in an increase in the pK_a for HNO_2.

(D) The volume used to measure the pH of sample 4 was less than 25.0 mL, resulting in a decrease in the concentration of $NO_2{}^-$.

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